EPSRC Centre for Doctoral Training in Sensor Technologies and Applications in an Uncertain World

Mini Research Project 1 Report guidelines

14/11/25 v0

1 **Introduction** 1
2 **Format** 1
2.1 Length and scope . . 1
2.2 Structure 1
2.3 Figures and Tables . . 2
3 **Advice for writing your research report** 2
3.1 Structure of research articles 2
3.2 Conventions, rules and styles 6
3.3 Order of writing . . . 7
4 **Assessment** 8

1 Introduction

The purpose of the report is twofold. Firstly, you will be demonstrating the research work you have accomplished during your project. Secondly, the report is an exercise in writing according to the standard conventions for your field. As this project is outside of your academic background, we are expecting you to learn many new skills, and so this will be taken into account during the assessment.

2 Format

2.1 Length and scope

The report should not exceed 7000 words, including the abstract, captions and appendices. However, conciseness is expected, and so 5000 words should be plenty sufficient for the main body of the report.

2.2 Structure

You should use the LaTeX template provided and use the accepted citation style for your field.

The recommended structure is:

- Title page (not included in word count)
- Abstract (strictly no more than 250 words)
- Main body
  - Introduction (around 500 words)
  - Methods (around 1500 words)
  - Results (around 2000 words)
  - Discussion (around 1000 words)
- References (not included in word count)

The word limit for the abstract is strict. The word limits for the other sections is indicative to help you know where your emphasis should be. Further advice about the structure of your report is given in 3.1.

2.3 Figures and Tables

You can include figures and tables. Make sure they are of good quality, legible, and numbered with an appropriate caption and citation if necessary. Even if you produce your own figure inspired by a published one, you should acknowledge this with 'Adapted from ...'.

3 Advice for writing your research report

3.1 Structure of research articles

Research articles generally contain these elements: Introduction, Methods, Results and Discussion. In many journals these are explicit section headings. In some journals this structure is implicit, but it is still being followed. The latter type of journals, e.g. Nature, often attach a detailed Methods section at the end of the paper or as a supplementary document. Nonetheless, the main text explains the experimental approach after the introductory paragraphs and before presenting results. Other variations exist to account for the needs of specific fields of study.

Across all variations on this theme, there is an important unifying feature: the breadth of discussion is hourglass shaped. The focus is wide at the beginning, narrows in the middle and opens up again at the end. Note that the final text rarely reflects the intermediate writing stages. Editing your first draft is the key to achieving this shape. If during the editing phase you find a generalising statement in the middle of the results section, simply integrate it into the discussion. Keeping this simple organising principle in mind will help you write a well-structured article.

On a second general note, novice writers typically spend most words on what took the most time to do. Usually, this

is not where the most interesting results are. The most instructive findings are often obtained at the end, which is where the emphasis of your paper should lie.

The four main sections have a finer substructure. If you analyse a few of your key references you are certain to find this overall pattern, even if not every paper addresses each point. The list has been adapted from Heard [1] p. 76:

[1]: Heard (2016), *The Scientist's Guide to Writing: How to Write More Easily and Effectively throughout Your Scientific Career*

- Introduction
  - General context
  - More specific research area and its importance
  - Identification of a gap, need, or contradiction
  - Specific research question to address this gap
  - Summary of approach to answer the question
  - Announcement of principle findings
- Methods
  - Materials, specimen, field sites, mathematical techniques
  - Observational/experimental procedures
  - Methods for data analysis
- Results
  - Results of observations, experiments or modelling (text, tables, figures)
  - Comparison among results (e.g. observation vs. theory)
- Discussion
  - Interpretation of results to answer research questions
  - Consideration of possible limitations
  - Relationship of results to previous literature
  - Broader implications having answered the research question
  - Prospects of future progress

3.1.1 Title

The title should raise the reader's interest and is the shortest summary of your work at 10 to 20 words. Try to address the questions "what?", "why?" and "how?". Scientific papers are not crime novels. You need reveal the culprit in the title to catch a scientist's interest: use keywords that predict the content; avoid acronyms and jargon. The title should reflect the tone of the paper.

3.1.2 Abstract

The abstract is a condensed version of your paper and reflects its structure. The motivation (1 to 2 sentences) and

problem statement (1 sentence) form your introduction, followed by the method, results and implications in sequence. Generally, abstracts are typeset as a single paragraph. The exception are medical journals, e.g. The Lancet, which use section headings and slightly longer abstracts. Reading some of these can give you guidance on the relative weighting of the abstract contents. The abstract should leave a reader with the interest to learn more, but not wondering what your paper might reveal if read. Do not use references, graphs or acronyms in the abstract.

3.1.3 Introduction

Your introduction orients the reader and sets the scene for your research. Your first few sentences will give the broadest context and often point at your underlying motivation. For example, if you worked on a solid-state device to measure particulate matter in air, you could start with figures on air pollution and its impact on health. This would be suitable for a generalist journal that covers many subjects, e.g. PNAS. In a journal of atmospheric science, most readers would be familiar with these numbers and you would start with something more specific, e.g. how a new type of material that you used enables more sensitive detection.

You then move to a concrete problem in your research area, for example by pointing out a gap or suggesting a new way of thinking. For example, applying single-molecule technology to a sensing problem allows the recording of a value's distribution rather than only the average.In the words of Heard [1], you establish a niche. Your next task is to occupy this niche, by outlining how you will apply your suggested way of thinking or answer a specific question. For example, you combined microfluidics and confocal imaging to study the dynamics of a disease-related protein at single-molecule resolution.

[1]: Heard (2016), *The Scientist's Guide to Writing: How to Write More Easily and Effectively throughout Your Scientific Career*

A good way to end an introduction is to announce your principal findings. This tells the reader where you are heading and thus helps to understand the motivations and design thinking behind the first set of experiments you present in the results section.

Please note that the introduction to your report is a distinct type of writing to the literature review, and that you should be careful not to self-plagiarise.

3.1.4 Methods

The methods section contains a detailed description of how you conducted your experiments. An often d principles is that another scientist should be able to repeat your experiments and obtain similar results. You might think that this makes the methods section the most objective. Yet, you do

have to make a subjective judgement on how much detail needs to be included. If you have tried to repeat an experiment from a published paper, you might already have found that making this judgement is difficult even for experienced authors. Word limits might indeed never allow every important detail to be included. In light of this, the deeper function of the methods section might be in establishing the credibility of your approach and its ability to deliver on the central research question.

Depending on your project this section could include:

- materials, e.g. a list of chemical compounds or the origin of samples you used.
- experimental setup and procedures
- data analysis, e.g. mathematical models you developed

Often you will use a procedure which has previously been published by the research group. There is no need to include a detailed description of this, you can the relevant study. Write this section in the past tense and avoid the use of the first person.

3.1.5 Results

In this section you present the outcomes of your research. The guiding principle is to present only the data that directly supports your answering of the research question and the telling of your story. A typical research paper will contain 4 to 5 display items, i.e. schematics, graphs and tables, while preliminary and supporting data is moved to appendices. So, again, you have to make an active judgement on what is central and what is secondary.

The methods, results and discussion aspects of an experiment seem naturally connected and you will feel tempted to use your individual experiments as the ordering principle, explaining as you go along. This is writing from your perspective. For the reader it is much more difficult to extract the important information from such a sequential text. Always try to write from the reader's perspective.

To aid a reader in understanding your results, think of your display items and text as telling the same story twice, i.e. the reader should be able to look only at your figures or only read the text and still understand the key results of the study. Of course, text and display items interact and enrich each other which deepens understanding. Therefore, the text should refer to all display items and point out what the reader is looking for, e.g. a trend or pattern. While looking for this trend, they should not need to refer back to the text. Each display item with its caption should contain all the information needed to understand it.

In the results section you present data, highlight trends and make comparisons, e.g. "The sensor was twice as sen-

sitive under condition x compared to condition y". The interpretation of such a result should be restricted to the discussion. In many journals the Results and Discussion are integrated and followed by a separate Conclusion. This can work well for a longer paper to reduce repetition. However, for this assessment you are asked separate the two, as this is a good exercise in raising your own awareness of what the difference in presenting and interpreting data is.

- Tables: datasets with many variables or when precise numbers are needed, not good for trends.
- Figures: to show trends and relationships, not good for many variables or precise numbers.

3.1.6 Discussion

You can think of the discussion as mirror image of your introduction. You start at a narrow focus, broaden out and re-address points made in the introduction.

You ended the introduction with an announcement of the principal findings, which you justified in your methods section and expanded on in your results section. You should now interpret these results and explore to what extent they answer your research question. While you highlighted the strengths of your approach in the introduction, you should include possible limitations of your study, but keep this short. You have now set the scene to relate your work to previous literature. You could consider if your study supports or contradicts current thinking in your field or if it opens new questions. Try to realistically appraise your outcomes. It is unlikely that you solved all the issues in your field, but it is equally unlikely that you made no contribution at all. Finally, consider directions for future work. Be specific, because "more research" is always necessary.

From an evaluator's perspective, this is section often best reveals how well you understood your project and how you think as a scientist. Are you able to realistically evaluate experimental results? Do you understand the literature well enough to put your findings into context? Are you able to plan future experiments based on what your results mean?

3.2 Conventions, rules and styles

This guidance has described writing according to conventions. Remember that conventions are not rules. Make sure you know what the rules are, e.g. the submission rules above, the requirements of a scientific journal or the indisputable prohibition of plagiarism. While you must stick to the rules, you can deviate from convention. When setting out as an academic writer, try to learn the conventions on report struc-

ture before deviating from them. It will give you the advantage of knowing what you do and why you do it.

Some conventions are less explicit. For example, there are some universal elements of style. These were explained beautifully in the classic text by Strunk and White [2]. Their considerations could be applied across all forms of writing, yet there seems to be an unwritten convention about recommending but ignoring them in academia. This was well expressed by Sword [3] p. 3:

[2]: Strunk et al. (1959), *The Elements of Style*

[3]: Sword (2012), *Stylish Academic Writing*

> Pick up a peer-reviewed journal in just about any academic discipline and what will you find? Impersonal, stodgy, jargon-laden, abstract prose that ignores or defies most of the stylistic principles [...]. There is a massive gap between what most readers consider to be good writing and what academics typically produce and publish. I'm not talking about the kinds of formal strictures necessarily imposed by journal editors — article length, citation style, and the like — but about a deeper, duller kind of disciplinary monotony, a compulsive proclivity for discursive obscurantism and circumambulatory diction (translation: an addiction to big words and soggy syntax).

You should realise that writing using "big words and soggy syntax" is not a rule and it is highly recommended to deviate from it. You can use a book on style, such as Alley [4], to improve your report once you are almost finished. Do not spend too much time on this, as your main focus remains the scientific content. Rather, try to improve your style over time with every piece of writing. It is beyond the scope of this document to give detailed guidance and examples, but it is best summarised by the ten statements on composition by Strunk and White [2]:

[4]: Alley (2018), *The Craft of Scientific Writing*

[2]: Strunk et al. (1959), *The Elements of Style*

1. Choose a suitable design [i.e. structure] and hold to it.
2. Make the paragraph the unit of composition.
3. Use the active voice.
4. Use definite, specific concrete language.
5. Omit needless words.
6. Avoid a succession of loose sentences.
7. Express coordinate ideas in similar form.
8. Keep related words together.
9. In summaries, keep to one tense.
10. Place the emphatic words of a sentence at the end.

3.3 Order of writing

As mentioned previously, the ordering of content in the final text is not reflective of the writing process. Ideally,

your process consists of the following for stages, with the amount of effort spent on each stage indicated in percentages:

First, you need to organise your contents. Write out in general bullet points what you want to include in each section. I would suggest working on the sections in the following sequence:

1. Results
2. Methods
3. Introduction and Discussion

Why this sequence? The results you present determine everything else: which methods you need to include to justify the results and what literature you need to cite to explain why the results matter.

Now write your first draft. This is a tricky stage, as you are not quite sure where you are heading and if your bullet point plan will make sense. Do not worry about these details now, do not stop yourself after each sentence to go back and edit. Allow your thoughts to flow and allow yourself to make mistakes. Once the first draft is in place you will have liberated yourself from many inhibiting doubts. You also simplified your task: critiquing and improving a concrete piece of writing is far easier than creating a new one.

Spend most of your time on the easier task of revising the first draft. The revision stage is about ensuring the logical flow and integrity of your content: move paragraphs around if needed, rework the sequence of sentences within a paragraph, add or remove a detail, add transition sentences between paragraphs, make sure you addressed the elements listed in 3.1, make sure you have the overall hour-glass shape.

Once you are satisfied with the main text, you can work on the abstract and title. The abstract is a summary of the main text, so it can only be written at the end. You might have a working title while you do the project, but your final title should be set at the end to reflect what you actually did. The reader should be able to understand your title, abstract and main text independently from each other.

Only once all of these elements are in place, should you spend time on editing, i.e. improving expression, style, grammar, orthography and punctuation of the whole paper.

4 Assessment

You should finish your laboratory work at the latest by the end of the MRes' Lent Term (i.e. Wednesday 18 March).

Your report should be uploaded to Moodle by Wednesday 25 March.

Assessment rubric

Criterion	Outstanding (≥85%)	Distinction (75–84%)	Pass (60–74%)	Unsatisfactory (<60%)
Background, Introduction and scope (10%)	Demonstrates authoritative command of the research domain, positioning the project at or near the current research frontier. Synthesises contemporary literature to identify a clear research gap, limitation, or unresolved challenge. The scope is precisely defined and justified with scholarly originality and strategic insight.	Demonstrates strong critical engagement with advanced literature and contemporary developments. The project is clearly situated within the field, and the scope is well defined, though research gaps are identified with limited depth or originality.	Demonstrates adequate familiarity with the field and relevant literature, but engagement is largely summarative. The scope is defined but weakly justified, with minimal critical positioning within current research.	Demonstrates insufficient understanding of the field. Literature use is sparse, outdated, or inappropriate. The project scope is ill-defined, uncritical, or disconnected from current research.
Understanding and analysis (30%)	Demonstrates exceptional methodological literacy and independent judgement. Methodology is critically designed or adapted, with explicit discussion of theoretical assumptions, constraints, and trade-offs. Analysis of outcomes is deeply analytical, reflexive, and evaluative, demonstrating research-level reasoning.	Demonstrates advanced understanding of methodology and clear justification of design choices. Limitations are critically discussed, though not fully problematised. Outcome analysis is rigorous but less reflexive or innovative.	Demonstrates functional understanding of methods, with limited justification and surface-level treatment of limitations. Outcome analysis is largely descriptive or procedural, showing limited synthesis.	Demonstrates weak or flawed understanding of methodology. Critical limitations are ignored or misunderstood. Analysis lacks rigour, coherence, or validity, falling below Masters level expectations.
Quality and originality of results (30%)	Produces technically robust, thoroughly validated results that demonstrate clear originality, independence, and potential publishability. Interpretation shows theoretical insight, critical comparison with existing work, and explicit articulation of contribution and significance.	Produces reliable and well-supported results that meet objectives and demonstrate elements of originality or advanced application. Interpretation is well reasoned but limited in theoretical or conceptual depth.	Produces results that meet objectives at a basic level but lack robustness, innovation, or depth. Interpretation is generally correct but under-developed and weakly contextualised.	Results are incomplete, unreliable, or poorly validated. Interpretation is incorrect, superficial, or absent. There is no discernible contribution or originality.
Presentation (30%)	The report demonstrates a professional, publication-ready standard. Argumentation is coherent and persuasive, writing is precise and academic, and figures are analytically informative, technically accurate, and seamlessly integrated. Presentation substantially enhances the work's impact.	The report is clearly structured and well written, with strong academic tone. Figures and tables are appropriate and clear, though some presentation choices limit analytical effectiveness.	The report is adequately structured but lacks clarity, polish, or argumentative coherence in places. Figures are present but may be unclear, weakly explained, or inconsistently formatted.	The report is poorly organised and difficult to follow. Writing lacks academic clarity. Figures are missing, misleading, or poorly integrated, significantly impairing comprehension.